BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
ResNet-50 Gender Classifier (crangana/trained-gender)

EVALUATION DATASET
FairFace Benchmark Dataset (Validation Split) (N=10954)

PROTECTED AXIS
Race

AUDIT DATE
2026-10-19 04:51:29 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples (*n < 30*) have point estimates suppressed to prevent unreliable compliance claims.

7 / 7	Overall (Table 1)	DIR = 0.720	0
SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05)	WORST DISPARITY COHORT (28 METRIC FAILURES)	WORST HEADLINE METRIC (OVERALL)	SUBGROUPS FLAGGED N < 30 (NFR-003)

EXECUTIVE TRANSLATION • NON-TECHNICAL REVIEW

FAIL

Demographic Disparity Flagged: Remediation & Review Required

The audit of ResNet-50 Gender Classifier (crangana/trained-gender) identified statistically verified demographic disparities affecting 7 of the 7 evaluated Race cohorts (N=10954 subjects, statistical significance $p < 0.05$). These performance gaps exceed random statistical fluctuation, indicating systematic differences in model predictions.

1 Disparity Impact

The highest demographic divergence occurred for Overall (Table 1), where the Disparate Impact Ratio (DIR) was 0.720. In plain terms, subjects in this group received positive predictions at only 72.0% the rate of the benchmark group—below the standard 80% legal threshold (a 28.0% shortfall). (Observed at the global cross-group level).

2 Underlying Drivers

When evaluating which characteristics most strongly correlate with the observed classification differences, surrogate Shapley attribution identified 'Gender (Male)', 'Gender (Female)', and 'Age (3-9)' as the primary demographic proxy factors. These reflect statistical associations within the dataset rather than direct physical causation.

3 Sample Reliability

All evaluated demographic cohorts met or exceeded the minimum sample size threshold ($n \geq 30$ subjects per NFR-003), ensuring statistical reliability across all reported metrics.

1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. Plotted on a calibrated [0.0, 1.0] domain with null-point reference and threshold lines.

▼ 📖 **How to Read These Metrics (In Plain English)**

Disparate Impact Ratio (DIR)

Selection Rate Ratio / 80% Rule

Compares the positive prediction rate of a demographic group against the highest-selected group. Under the US EEOC 4/5ths (80%) rule, if the highest group is selected 80% of the time, all other groups should be selected at least 64% of the time (80% of 80%). A ratio below 0.80 indicates adverse impact.

`Ideal = 1.00; Safe threshold ≥ 0.80 • EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)`

Demographic Parity Difference (DPD)

Positive Rate Percentage Gap

Measures whether the model favors one group over another in raw volume of positive predictions. A difference of 0.15 means individuals from one group are 15 percentage points more likely to receive a positive prediction than individuals from another group.

`Ideal = 0.00; Safe threshold ≤ 0.10 • EU AI Act Art. 10(2)(f) / NIST AI RMF Measure 2.11`

Equalized Odds Difference (EOD)

Error Rate Equality Gap

Ensures the model is equally accurate for all demographics. It verifies that qualified individuals have an equal chance of being accepted, and unqualified individuals have an equal chance of being rejected, regardless of identity.

`Ideal = 0.00; Safe threshold ≤ 0.10 • EU AI Act Art. 10(2)(g) / NIST AI RMF Measure 2.11`

Equal Opportunity Difference (EOP)

True Positive Opportunity Gap

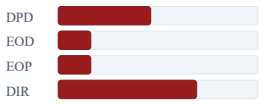
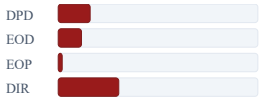
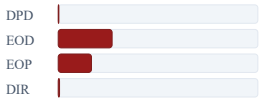
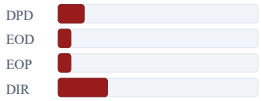
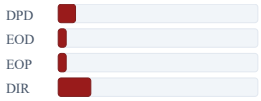
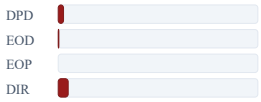
Focuses strictly on true positive outcomes (e.g. qualified applicants or correct identity matches). A difference of 0.08 means qualified individuals in one group are 8 percentage points less likely to be correctly recognized.

`Ideal = 0.00; Safe threshold ≤ 0.10 • EU AI Act Art. 10(3) / NIST AI RMF Measure 2.11`

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null	Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.187	[0.144, 0.214]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	0.118	[0.091, 0.141]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	0.100	[0.076, 0.130]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.720	[0.687, 0.774]		p=0.0000 **p < 0.05**	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = p < 0.05)	Data Integrity Status
Middle Eastern	1209	0.139	0.049	0.049	0.792		Audited (Valid)
Latino_Hispanic	1623	0.048	0.035	0.006	0.909		Audited (Valid)
Black	1556	0.001	0.081	0.050	0.998		Audited (Valid)
East Asian	1550	0.039	0.019	0.019	0.926		Audited (Valid)
Indian	1516	0.026	0.012	0.012	0.951		Audited (Valid)
White	2085	0.008	0.001	0.000	0.985		Audited (Valid)
Southeast Asian	1415	0.006	0.002	0.002	0.988		Audited (Valid)

3. Targeted Disparity Explainability & Proxy Attributions

Demographic Surrogate Associations — Mean absolute contribution to surrogate log-odds

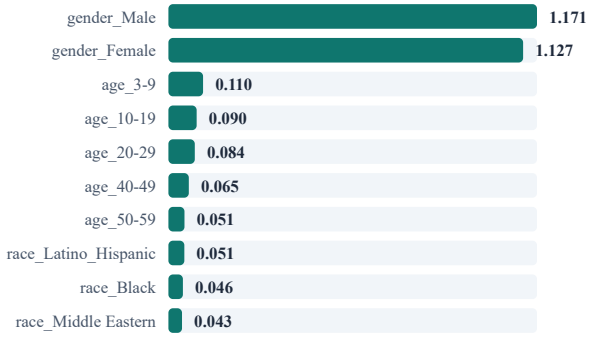
Surrogate additive Shapley attributions computed for statistically flagged disparities ($p < 0.05$, $n \geq 30$). A global ridge-regularized logistic surrogate model is fitted to predictions across the full evaluation dataset to estimate feature proxy influence. Mean absolute Shapley attributions measure the proxy contribution of each demographic feature to surrogate log-odds.

> **Non-Causal Disclaimer (Bilodeau et al., 2024):** Attributions identify empirical associations between demographic proxy features and model predictions. They do *not* establish causal relationships or prescribe remediation. This is a diagnostic tool only — no model retraining, in-situ weight modification, or causal intervention is performed.

Statistically Flagged Disparities Triggering Surrogate Analysis

race=Global • DPD

Global Demographic Proxy Feature Importance



Demographic Feature	Mean absolute contribution to surrogate log-odds
gender_Male	1.1714
gender_Female	1.1273
age_3-9	0.1101
age_10-19	0.0897
age_20-29	0.0838
age_40-49	0.0648
age_50-59	0.0510
race_Latino_Hispanic	0.0506
race_Black	0.0460
race_Middle Eastern	0.0432

4. Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-1k pretrained ResNet-50 feature extractor fine-tuned on the FairFace balanced demographic training split, evaluating baseline residual scaling on subgroup disparity.

Backend configuration: dual. Both backend executions completed.

Cross-backend agreement checks point estimates only; bootstrap, hypothesis tests, and reporting share implementation. Consult execution logs for backend failures and divergences.